

JEE MAIN REVISION MASTER: PROPERTIES OF TRIANGLE OUT OF SYLLABUS

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 36 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Properties Of Triangle — Standard Calculations & Problem Archetypes	20 PYQs	Quantitative Analysis
High	Properties Of Triangle — Core Principles & Governing Laws	8 PYQs	Core Principle
High	Properties Of Triangle — Applied Cases, Reactions & Traps	8 PYQs	Applied Topics

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Properties Of Triangle — Standard Calculations & Problem Archetypes	$r=3, \quad \frac{R}{r}, \quad \angle BAC=90^\circ$
Properties Of Triangle — Core Principles & Governing Laws	$\angle C=\frac{\pi}{2}, \quad \rightarrow c=2R, \quad \sqrt{3}(BE)=4(AB)$
Properties Of Triangle — Applied Cases, Reactions & Traps	$\cos B=\frac{3}{5} \rightarrow B=53^\circ, \quad r+R=\frac{a^2}{\cot \frac{\pi}{n}+\operatorname{cosec} \frac{\pi}{n}}, \quad =\frac{a^2}{\frac{\cos \frac{\pi}{n}+1}{\sin \frac{\pi}{n}}}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Properties Of Triangle — Standard Calculations & Problem Archetypes (Very High Yield — 20 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Properties Of Triangle.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2025 Morning Shift] Q#1 (MCQ): "Let the area of a $\triangle PQR$ with vertices $P(5,4), Q(-2,4)$ and $R(a,b)$ be 35 square units. If its orthocenter and centroid are $O(2, \frac{14}{3})$..." [Ans: N/A]
- [2024 Evening Shift] Q#1 (Numerical): "Consider a triangle ABC having the vertices $A(1,2), B(\alpha, \beta)$ and $C(\gamma, \delta)$ and angles $\angle ABC = (\pi/6)$ and $\angle \dots$ " [Ans: 14]
- [2024 Evening Shift] Q#2 (MCQ): "Two vertices of a triangle ABC are $A(3,-1)$ and $B(-2,3)$, and its orthocentre is $P(1,1)$. If the coordinates of the point C are $(\dots)$ " [Ans: N/A]

3.2 Properties Of Triangle — Core Principles & Governing Laws (High Yield — 8 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Properties Of Triangle.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Properties Of Triangle.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Properties Of Triangle PYQs.

Representative Source PYQs:

- [2024 Morning Shift] Q#3 (MCQ): "Let $(5, (a/4))$ be the circumcenter of a triangle with vertices $A(a, -2), B(a, 6)$ and $C((a/4), -2)$. Let α denote the cir..." [Ans: N/A]
- [2023 Evening Shift] Q#3 (Numerical): "In the figure, $\theta_1 + \theta_2 = (\pi/2)$ and $\sqrt{3}(BE) = 4(AB)$. If the area of $\triangle CAB$ is $2\sqrt{3} - 3$ unit², when $\dots$ " [Ans: 6]
- [2022 Morning Shift] Q#3 (MCQ): "The lengths of the sides of a triangle are $10 + x^2$, $10 + x^2$ and $20 - 2x^2$. If for $x = k$, the area of the triangle is maximum, then $3k \dots$ " [Ans: N/A]

3.3 Properties Of Triangle — Applied Cases, Reactions & Traps (High Yield — 8 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Properties Of Triangle.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2023 Morning Shift] Q#2 (MCQ): "A straight line cuts off the intercepts $OA=a$ and $OB=b$ on the positive directions of x -axis and y axis respectively. If the perpendi..." [Ans: N/A]
- [2023 Morning Shift] Q#2 (Numerical): "If the line $x=y=z$ intersects the line $x\sin A + y\sin B + z\sin C - 18 = 0 = x\sin 2A + y\sin 2B + z\sin 2C - 9$, where A, B, C are the angles of a triangle ABC , ..." [Ans: 5]
- [2021 Morning Shift] Q#2 (MCQ): "If in a triangle ABC , $AB = 5$ units, $\angle B = \cos^{-1}((3/5))$ and radius of circumcircle of $\triangle ABC$ is 5 units, then the area (in..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2025 (Online) 23rd January Morning Shift (Q#1)

Let the area of a $\triangle PQR$ with vertices $P(5,4)$, $Q(-2,4)$ and $R(a,b)$ be 35 square units. If its orthocenter and centroid are $O(2, (14/5))$ and $C(c,d)$ respectively, then $c+2d$ is equal to

- (A) 3
- (B) $(7/3)$
- (C) 2
- (D) $(8/3)$

Correct Answer: None

Solution Analysis: Equation of line $QR = 5x + 2y + 2 = 0$ Equation of line $PR = 10x - 3y - 38 = 0 \therefore$ Point $R(2, -6)$ Centroid $= ((5 + (-2) + a)/3, (4 + 4 + b)/3) = ((5/3), (2/3))$ $c + 2d = (5/3) + (4/3) = 3$

Benchmark Problem 2: JEE Main 2024 (Online) 29th January Morning Shift (Q#3)

Let $(5, (a/4))$ be the circumcenter of a triangle with vertices $A(a, -2)$, $B(a, 6)$ and $C((a/4), -2)$. Let α denote the circumradius, β denote the area and γ denote the perimeter of the triangle. Then $\alpha + \beta + \gamma$ is

- (A) 60
- (B) 62
- (C) 53
- (D) 30

Correct Answer: None

Solution Analysis: $A(a, -2), B(a, 6), C((a/4), -2), O(5, (a/4))$ $AO = BO \Rightarrow (a-5)^2 + ((a/4)+2)^2 = (a-5)^2 + ((a/4)-6)^2 \Rightarrow a = 8$
 $AB = 8, AC = 6, BC = 10 \Rightarrow \alpha = 5, \beta = 24, \gamma = 24$

Benchmark Problem 3: JEE Main 2023 (Online) 30th January Morning Shift (Q#2)

A straight line cuts off the intercepts $OA = a$ and $OB = b$ on the positive directions of x -axis and y -axis respectively. If the perpendicular from origin O to this line makes an angle of $(\pi/6)$ with positive direction of y -axis and the area of $\triangle OAB$ is $(98/3)\sqrt{3}$, then $a^2 - b^2$ is equal to :

- (A) $(392/3)$
- (B) 98
- (C) 196
- (D) $(196/3)$

Correct Answer: None

Solution Analysis: $(1/2)ab = \frac{98\sqrt{3}}{3} \Rightarrow \sqrt{3}ab = 196 \dots (i)$ $OP = OB \cos 30^\circ = OA \cos 60^\circ \Rightarrow \frac{b\sqrt{3}}{2} = \frac{a}{2} \Rightarrow \sqrt{3}b = a \dots (ii)$ By (i) and (ii) $a^2 = 196 \Rightarrow a = 14$ $b^2 = \frac{a^2}{3} \Rightarrow a^2 - b^2 = \frac{2a^2}{3} = \frac{392}{3}$